

Process & Decision Documentation

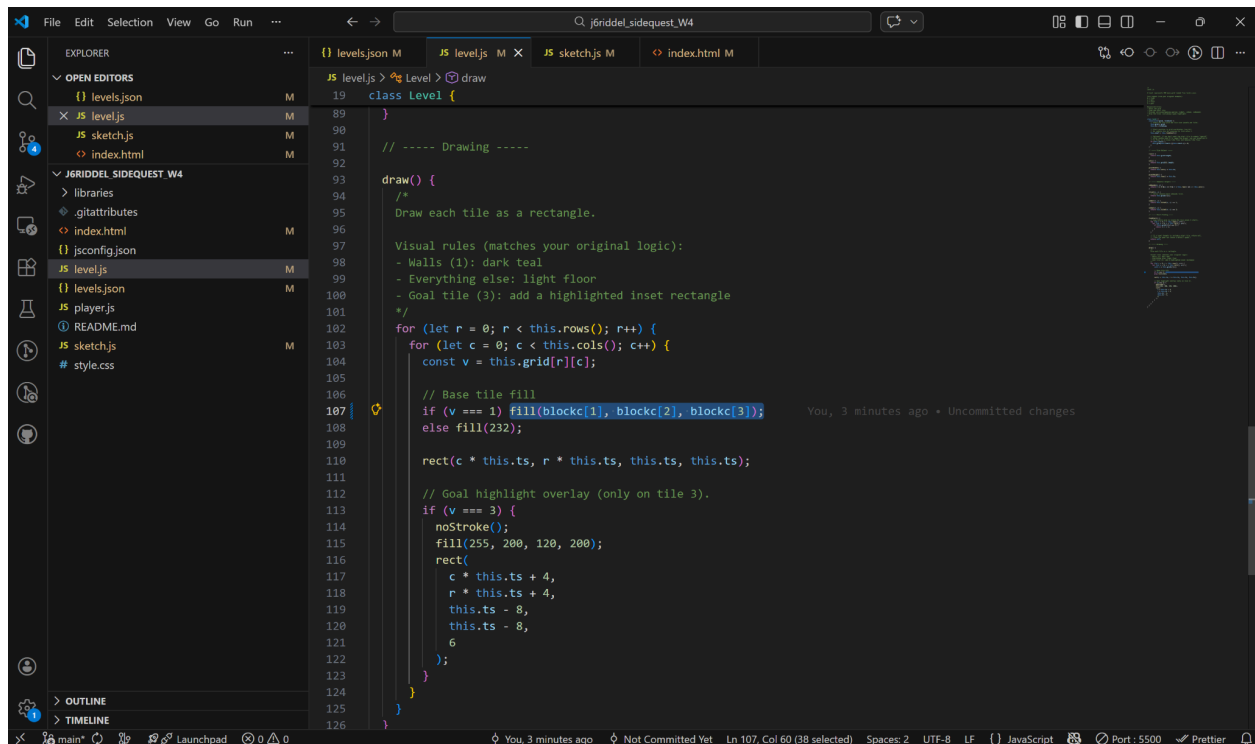
Project/Assignment Decisions

Side Quest Week 3

I used a json file to add another level and changed the colour of the level based on the level indicator using an array.

Role-Based Process Evidence

Colour pulled from arrays if correct block category from json

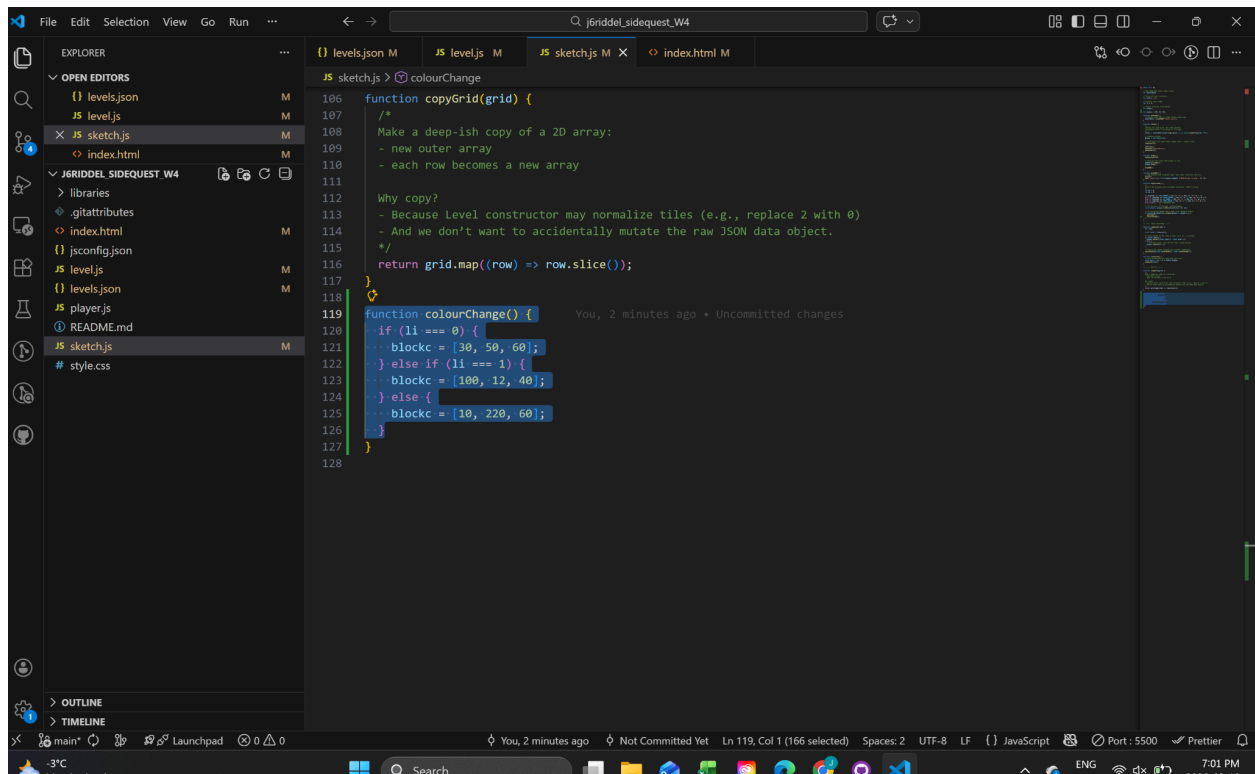


```
File Edit Selection View Go Run ...
j6ridde1.sidequest.W4
() levels.json M JS levels.js M X JS sketch.js M index.html M
EXPLORED
OPEN EDITORS
() levels.json M
X JS levels.js M
JS sketch.js M
index.html M
J6RIDDEL SIDEQUEST W4
libraries
.gitattributes
index.html M
() jsconfig.json
JS levels.js M
() levels.json M
JS player.js
README.md
JS sketch.js
style.css
OUTLINE
TIMELINE
main* Launchpad 0 0 0
You, 3 minutes ago • Not Committed Yet Ln 107, Col 60 (38 selected) Spaces: 2 UTF-8 LF JavaScript Port: 5500 Prettier

() levels.json M
JS levels.js M X
JS sketch.js M
index.html M

JS levels.js > Level > draw
19 class Level {
20 }
21
22 // ----- Drawing -----
23
24 draw() {
25   /*
26    Draw each tile as a rectangle.
27
28    Visual rules (matches your original logic):
29    - Walls (1): dark teal
30    - Everything else: light floor
31    - Goal tile (3): add a highlighted inset rectangle
32    */
33   for (let r = 0; r < this.rows(); r++) {
34     for (let c = 0; c < this.cols(); c++) {
35       const v = this.grid[r][c];
36
37       // Base tile fill
38       if (v === 1) fill(blockc[1], blockc[2], blockc[3]);
39       else fill(232);
40
41       rect(c * this.ts, r * this.ts, this.ts, this.ts);
42
43       // Goal highlight overlay (only on tile 3).
44       if (v === 3) {
45         noStroke();
46         fill(255, 200, 120, 200);
47         rect(
48           c * this.ts + 4,
49           r * this.ts + 4,
50           this.ts - 8,
51           this.ts - 8,
52           6
53         );
54       }
55     }
56   }
57 }
```

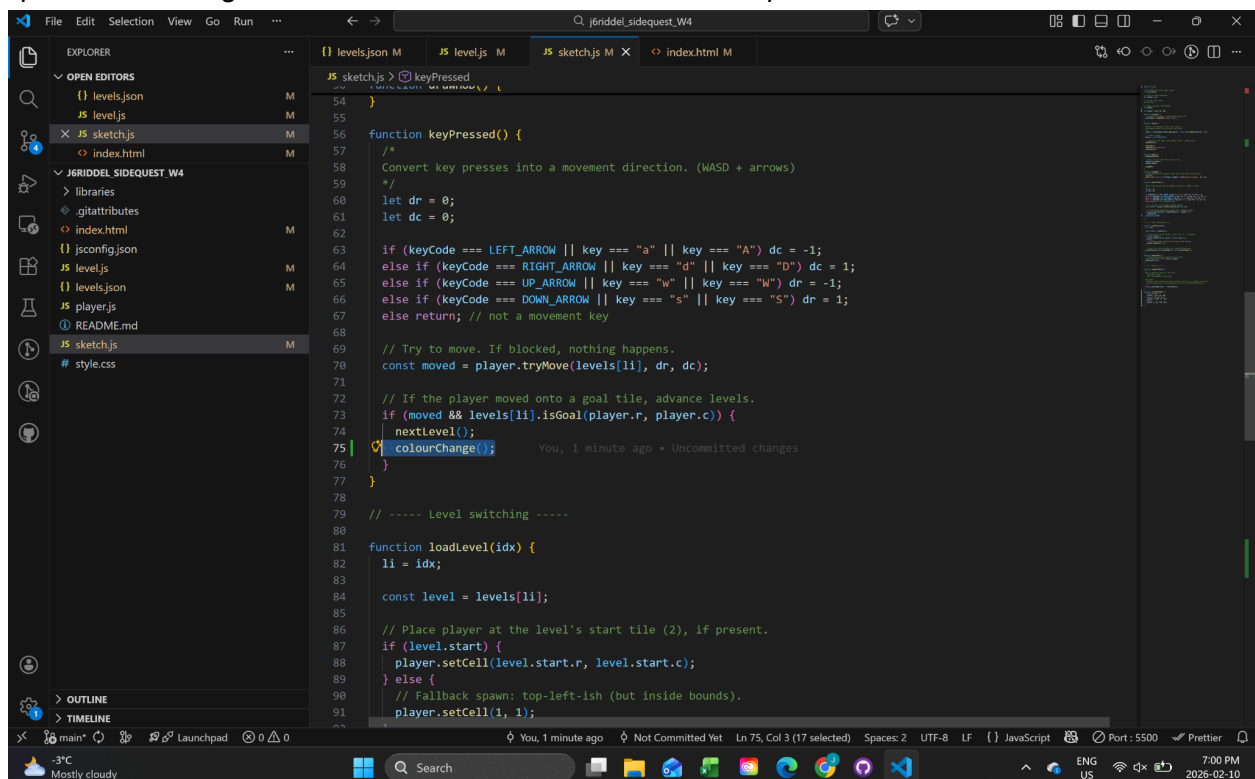
Function to change array and define color variable "blockc"



The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor window shows the 'sketch.js' file with the 'colourChange' function. The function is designed to update the 'blockc' variable based on the current level completion. It includes a comment explaining the purpose of the function and the logic for updating the array and color variable.

```
106 function copyGrid(grid) {
107   /*
108    * Make a deep-ish copy of a 2D array:
109    * - new outer array
110    * - each row becomes a new array
111    */
112   Why copy?
113   - Because level constructor may normalize tiles (e.g., replace 2 with 0)
114   - And we don't want to accidentally mutate the raw JSON data object.
115   */
116   return grid.map((row) => row.slice());
117 }
118
119 function colourChange() {
120   if (li === 0) {
121     blockc = [30, 50, 60];
122   } else if (li === 1) {
123     blockc = [100, 12, 40];
124   } else {
125     blockc = [10, 220, 60];
126   }
127 }
128
```

I put colour change in the if statement that check level completion



The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor window shows the 'sketch.js' file with the 'keyPressed' function. The function is designed to handle key presses and update the game state. It includes a comment explaining the purpose of the function and the logic for updating the game state, including the 'colourChange' function call.

```
54 }
55
56 function keyPressed() {
57   /*
58    * Convert key presses into a movement direction. (WASD + arrows)
59    */
60   let dr = 0;
61   let dc = 0;
62
63   if (keyCode === LEFT_ARROW || key === "a" || key === "A") dc = -1;
64   else if (keyCode === RIGHT_ARROW || key === "d" || key === "D") dc = 1;
65   else if (keyCode === UP_ARROW || key === "w" || key === "W") dr = -1;
66   else if (keyCode === DOWN_ARROW || key === "s" || key === "S") dr = 1;
67   else return; // not a movement key
68
69   // Try to move. If blocked, nothing happens.
70   const moved = player.tryMove(levels[li], dr, dc);
71
72   // If the player moved onto a goal tile, advance levels.
73   if (moved && levels[li].isGoal(player.r, player.c)) {
74     nextLevel();
75     colourChange();
76   }
77 }
78
79 // ----- Level switching -----
80
81 function loadLevel(idx) {
82   li = idx;
83
84   const level = levels[li];
85
86   // Place player at the level's start tile (2), if present.
87   if (level.start) {
88     player.setCell(level.start.r, level.start.c);
89   } else {
90     // Fallback spawn: top-left-ish (but inside bounds).
91     player.setCell(1, 1);
92   }
93 }
```

Initialized blockc

The screenshot shows the VS Code editor with the file explorer on the left. The 'EXPLORER' panel shows the project structure for 'j6ridde1.sidequest.W4'. The 'OPEN EDITORS' panel shows the following files: levels.json, JS levels.js, JS sketch.js, and index.html. The 'JS sketch.js' file is open in the editor. The code in the editor is as follows:

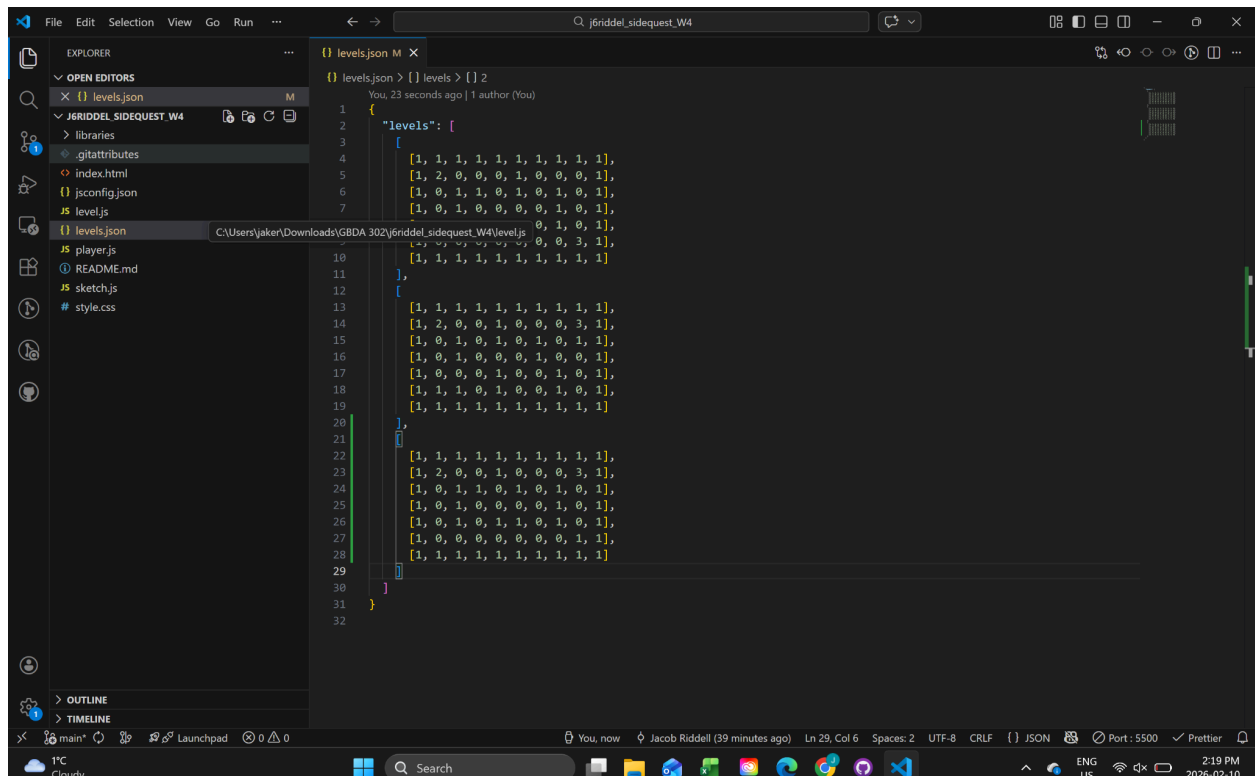
```
1  const TS = 32;
2
3  // Raw JSON data (from levels.json).
4  let levelsData;
5
6  // Array of Level instances.
7  let levels = [];
8
9  // Current level index.
10 let li = 0;
11
12 // Player instance (tile-based).
13 let player;
14
15 let blockc = [50, 50, 50];
16
17 function preload() {
18   // Ensure level data is ready before setup runs.
19   levelsData = loadJSON("levels.json");
20 }
21
22 function setup() {
23   /*
24    Convert raw JSON grids into Level objects.
25    levelsData.levels is an array of 2D arrays.
26    */
27   levels = levelsData.levels.map((grid) => new Level(copyGrid(grid), TS));
28
29   // Create a player.
30   player = new Player(TS);
31
32   // Load the first level (sets player start + canvas size).
33   loadLevel(0);
34
35   noStroke();
36   textFont("sans-serif");
37   textSize(14);
38 }
```

Failed attempt at using random colours in if

The screenshot shows the VS Code editor with the file explorer on the left. The 'EXPLORER' panel shows the project structure for 'j6ridde1.sidequest.W4'. The 'OPEN EDITORS' panel shows the following files: levels.json, JS levels.js, JS sketch.js, and index.html. The 'JS sketch.js' file is open in the editor. The code in the editor is as follows:

```
39 function draw() {
40   drawHUD();
41 }
42
43 function drawHUD() {
44   // HUD matches your original idea: show level count and controls.
45   fill(0);
46   text('Level ${(li + 1)}/${levels.length} - WASD/Arrows to move', 10, 16);
47 }
48
49 function keyPressed() {
50   /*
51    Convert key presses into a movement direction. (WASD + arrows)
52    */
53   let dr = 0;
54   let dc = 0;
55
56   if (keyCode === LEFT_ARROW || key === "a" || key === "A") dc = -1;
57   else if (keyCode === RIGHT_ARROW || key === "d" || key === "D") dc = 1;
58   else if (keyCode === UP_ARROW || key === "w" || key === "W") dr = -1;
59   else if (keyCode === DOWN_ARROW || key === "s" || key === "S") dr = 1;
60   else return; // not a movement key
61
62   // Try to move. If blocked, nothing happens.
63   const moved = player.tryMove(levels[li], dr, dc);
64
65   // If the player moved onto a goal tile, advance levels.
66   if (moved && levels[li].isGoal(player.r, player.c)) {
67     nextLevel();
68     fill(random(0, 225), random(0, 255), random(0, 255));
69   }
70
71   // ----- Level switching -----
72
73   function loadLevel(idx) {
74     li = idx;
75   }
76 }
```

Added third level



```
1 {
2   "levels": [
3     [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
4     [1, 2, 0, 0, 0, 1, 0, 0, 0, 1],
5     [1, 0, 1, 1, 0, 1, 0, 1, 0, 1],
6     [1, 0, 1, 0, 0, 0, 0, 1, 0, 1],
7     [1, 0, 1, 0, 0, 0, 0, 1, 0, 1],
8     [1, 0, 1, 0, 0, 0, 0, 1, 0, 1],
9     [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
10    ],
11  },
12  [
13    [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
14    [1, 2, 0, 0, 1, 0, 0, 0, 3, 1],
15    [1, 0, 1, 0, 1, 0, 1, 0, 1, 1],
16    [1, 0, 1, 0, 0, 0, 1, 0, 0, 1],
17    [1, 0, 0, 0, 1, 0, 0, 1, 0, 1],
18    [1, 1, 1, 0, 1, 0, 0, 1, 0, 1],
19    [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
20  ],
21  },
22  [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
23  [1, 2, 0, 0, 1, 0, 0, 0, 3, 1],
24  [1, 0, 1, 1, 0, 1, 0, 1, 0, 1],
25  [1, 0, 1, 0, 0, 0, 0, 1, 0, 1],
26  [1, 0, 1, 0, 1, 1, 0, 1, 0, 1],
27  [1, 0, 0, 0, 0, 0, 0, 0, 1, 1],
28  [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
29  ],
30  ],
31  }
32 }
```

Entry Header

Name: Jake Riddell

Role: Programmer

Goal of Work Session

I was learning how to change game state based off of level state and java script object notation.

Tools, Resources, or Inputs Used

- <https://p5js.org/reference/>
- p5.js
- Vs code
- Github
- GBDA302 Week 4 template 4

GenAI Documentation

No GenAI use